MySQL 8.0 Command Line Cli

```
Enter password: *************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| employee           |
| information_schema |
| librarydb          |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.00 sec)

mysql> use employee;
Database changed
mysql> select * from Employee;
ERROR 1146 (42S02): Table 'employee.employee' doesn't exist
mysql> select * from Dept;
ERROR 1146 (42S02): Table 'employee.dept' doesn't exist
mysql> select * from Emp;
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |
|   112 | Arjun   | 63800.00 |     10 | 2025-06-13  |
+-------+---------+----------+--------+-------------+
7 rows in set (0.00 sec)

mysql> create table Dept (DeptId int primary key , DeptName varchar(50), Location vachar(50));
```

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```
mysql> create table Dept (DeptId int primary key , DeptName varchar(50), Location vachar(50));
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'v
achar(50))' at line 1
mysql> create table Dept (DeptId int primary key , DeptName varchar(50), Location varchar(50));
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Dept (DeptID, DeptName, Location)
    -> VALUES
    -> (10, 'IT', 'Mumbai'),
    -> (20, 'HR', 'Pune'),
    -> (30, 'Sales', 'Mumbai'),
    -> (40, 'Finance', 'Delhi'),
    -> (50, 'Marketing', 'Nashik');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from Dept;
+--------+-----------+----------+
| DeptId | DeptName  | Location |
+--------+-----------+----------+
|     10 | IT        | Mumbai   |
|     20 | HR        | Pune     |
|     30 | Sales     | Mumbai   |
|     40 | Finance   | Delhi    |
|     50 | Marketing | Nashik   |
+--------+-----------+----------+
5 rows in set (0.00 sec)

mysql> create table Project(ProjectID int primary key, ProjectName varchar(100), DeptID int);
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Project (ProjectID, ProjectName, DeptID)
    -> VALUES
    -> (1, 'Website Development', 10),
    -> (2, 'Recruitment System', 20),
    -> (3, 'Sales Dashboard', 30),
    -> (4, 'Accounting System', 40),
    -> (5, 'Marketing Campaign', 50);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from Project;
+-----------+--------------------+--------+
| ProjectID | ProjectName        | DeptID |
+-----------+--------------------+--------+
|         1 | Website Development |     10 |
```

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```
mysql> select * from Project;
+-----------+---------------------+--------+
| ProjectID | ProjectName         | DeptID |
+-----------+---------------------+--------+
|         1 | Website Development |     10 |
|         2 | Recruitment System  |     20 |
|         3 | Sales Dashboard     |     30 |
|         4 | Accounting System   |     40 |
|         5 | Marketing Campaign  |     50 |
+-----------+---------------------+--------+
5 rows in set (0.00 sec)

mysql> select * from Emp;
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |
|   112 | Arjun   | 63800.00 |     10 | 2025-06-13  |
+-------+---------+----------+--------+-------------+
7 rows in set (0.00 sec)

mysql> Emp × Dept
    -> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'E
mp × Dept' at line 1
mysql> select Emp.EmpName , Dept.DeptName from Emp inner join Dept on Emp.DeptID = Dept.DeptID;
+---------+----------+
| EmpName | DeptName |
+---------+----------+
| Sakshi  | IT       |
| Shruti  | HR       |
| Sumit   | Sales    |
| Pranav  | IT       |
| Om      | HR       |
| Malhar  | Sales    |
| Arjun   | IT       |
+---------+----------+
7 rows in set (0.00 sec)

mysql> select Emp.EmpName , Dept.DeptName from Emp left join Dept on Emp.DeptID = Dept.DeptID;
+---------+----------+
| EmpName | DeptName |
```

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```
mysql> select Emp.EmpName , Dept.DeptName from Emp left join Dept on Emp.DeptID = Dept.DeptID;
+---------+----------+
| EmpName | DeptName |
+---------+----------+
| Sakshi  | IT       |
| Shruti  | HR       |
| Sumit   | Sales    |
| Pranav  | IT       |
| Om      | HR       |
| Malhar  | Sales    |
| Arjun   | IT       |
+---------+----------+
7 rows in set (0.00 sec)

mysql> select Emp.EmpName , Dept.DeptName from Emp right join Dept on Emp.DeptID = Dept.DeptID;
+---------+-----------+
| EmpName | DeptName  |
+---------+-----------+
| Sakshi  | IT        |
| Pranav  | IT        |
| Arjun   | IT        |
| Shruti  | HR        |
| Om      | HR        |
| Sumit   | Sales     |
| Malhar  | Sales     |
| NULL    | Finance   |
| NULL    | Marketing |
+---------+-----------+
9 rows in set (0.00 sec)

mysql> select * from Emp natural join Dept;
+--------+-------+---------+----------+-------------+----------+----------+
| DeptID | EmpID | EmpName | Salary   | JoiningDate | DeptName | Location |
+--------+-------+---------+----------+-------------+----------+----------+
|     10 |   101 | Sakshi  | 88000.00 | 2023-01-25  | IT       | Mumbai   |
|     20 |   102 | Shruti  | 55000.00 | 2023-11-12  | HR       | Pune     |
|     30 |   103 | Sumit   | 70000.00 | 2022-10-21  | Sales    | Mumbai   |
|     10 |   104 | Pranav  | 82500.00 | 2025-11-14  | IT       | Mumbai   |
|     20 |   105 | Om      | 50000.00 | 2020-10-11  | HR       | Pune     |
|     30 |   106 | Malhar  | 65000.00 | 2022-09-09  | Sales    | Mumbai   |
|     10 |   112 | Arjun   | 63800.00 | 2025-06-13  | IT       | Mumbai   |
+--------+-------+---------+----------+-------------+----------+----------+
7 rows in set (0.00 sec)

mysql> select * from Emp cross join Project;
+-------+---------+----------+--------+-------------+-----------+--------------------+--------+
```

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```
mysql> select * from Emp cross join Project;
+-------+---------+----------+--------+-------------+-----------+---------------------+--------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate | ProjectID | ProjectName         | DeptID |
+-------+---------+----------+--------+-------------+-----------+---------------------+--------+
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |         5 | Marketing Campaign  |     50 |
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |         4 | Accounting System   |     40 |
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |         3 | Sales Dashboard     |     30 |
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |         2 | Recruitment System  |     20 |
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |         1 | Website Development |     10 |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |         5 | Marketing Campaign  |     50 |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |         4 | Accounting System   |     40 |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |         3 | Sales Dashboard     |     30 |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |         2 | Recruitment System  |     20 |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |         1 | Website Development |     10 |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |         5 | Marketing Campaign  |     50 |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |         4 | Accounting System   |     40 |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |         3 | Sales Dashboard     |     30 |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |         2 | Recruitment System  |     20 |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |         1 | Website Development |     10 |
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |         5 | Marketing Campaign  |     50 |
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |         4 | Accounting System   |     40 |
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |         3 | Sales Dashboard     |     30 |
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |         2 | Recruitment System  |     20 |
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |         1 | Website Development |     10 |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |         5 | Marketing Campaign  |     50 |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |         4 | Accounting System   |     40 |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |         3 | Sales Dashboard     |     30 |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |         2 | Recruitment System  |     20 |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |         1 | Website Development |     10 |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |         5 | Marketing Campaign  |     50 |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |         4 | Accounting System   |     40 |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |         3 | Sales Dashboard     |     30 |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |         2 | Recruitment System  |     20 |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |         1 | Website Development |     10 |
|   112 | Arjun   | 63800.00 |     10 | 2025-06-13  |         5 | Marketing Campaign  |     50 |
|   112 | Arjun   | 63800.00 |     10 | 2025-06-13  |         4 | Accounting System   |     40 |
|   112 | Arjun   | 63800.00 |     10 | 2025-06-13  |         3 | Sales Dashboard     |     30 |
|   112 | Arjun   | 63800.00 |     10 | 2025-06-13  |         2 | Recruitment System  |     20 |
|   112 | Arjun   | 63800.00 |     10 | 2025-06-13  |         1 | Website Development |     10 |
+-------+---------+----------+--------+-------------+-----------+---------------------+--------+
35 rows in set (0.00 sec)

mysql> create table EmpBackup like Emp;
Query OK, 0 rows affected (0.07 sec)

mysql> insert into EmpBackup select * from Emp;
Query OK, 7 rows affected (0.01 sec)
```

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```
mysql> select * from EmpBackup;
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |
|   112 | Arjun   | 63800.00 |     10 | 2025-06-13  |
+-------+---------+----------+--------+-------------+
7 rows in set (0.00 sec)

mysql> create table ProjectArchive as select * from Project;
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> create table Emp_Auto(EmpID int Auto_Increment primary key, EmpName varchar(50), Salary decimal(10,2), DeptID int, JoiningDate Date);
Query OK, 0 rows affected (0.03 sec)

mysql> insert into Emp_Auto(EmpName,Salary, DeptID, JoiningDate) values ('Anita',50000,10,'2025-01-10');
Query OK, 1 row affected (0.02 sec)

mysql> select * from Emp_Auto;
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|     1 | Anita   | 50000.00 |     10 | 2025-01-10  |
+-------+---------+----------+--------+-------------+
1 row in set (0.00 sec)

mysql> select * from Emp where Salary > (select AVG(Salary) from Emp);
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |
+-------+---------+----------+--------+-------------+
3 rows in set (0.01 sec)

mysql> select * from Emp where DeptID in(select DeptID from Dept where location = 'Mumbai');
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |
```

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```
mysql> select * from Emp where DeptID in(select DeptID from Dept where location = 'Mumbai');
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |
|   112 | Arjun   | 63800.00 |     10 | 2025-06-13  |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |
+-------+---------+----------+--------+-------------+
5 rows in set (0.00 sec)

mysql> select * from Dept where DeptID in (select DeptID from Emp group by DeptID having count(*) > 5);
Empty set (0.01 sec)

mysql> select *  from Emp where salary = (select max(Salary) from Emp);
+-------+---------+----------+--------+-------------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate |
+-------+---------+----------+--------+-------------+
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |
+-------+---------+----------+--------+-------------+
1 row in set (0.00 sec)

mysql> select max(Salary) as MaximumSalary from Emp;
+---------------+
| MaximumSalary |
+---------------+
|      88000.00 |
+---------------+
1 row in set (0.00 sec)

mysql> select E.EmpName as Employee, M.EmpName as Manager from Emp E left join Emp M on E.ManagerID = M.EmpID;
ERROR 1054 (42S22): Unknown column 'E.ManagerID' in 'on clause'
mysql> alter table Emp  add ManagerID int;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> UPDATE Emp
    -> SET ManagerID = 101
    -> WHERE EmpID IN (102, 103, 104, 105);
Query OK, 4 rows affected (0.01 sec)
Rows matched: 4  Changed: 4  Warnings: 0

mysql> select * from Emp;
+-------+---------+----------+--------+-------------+-----------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate | ManagerID |
```

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```
+-------+---------+----------+--------+-------------+-----------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate | ManagerID |
+-------+---------+----------+--------+-------------+-----------+
|   101 | Sakshi  | 88000.00 |     10 | 2023-01-25  |      NULL |
|   102 | Shruti  | 55000.00 |     20 | 2023-11-12  |       101 |
|   103 | Sumit   | 70000.00 |     30 | 2022-10-21  |       101 |
|   104 | Pranav  | 82500.00 |     10 | 2025-11-14  |       101 |
|   105 | Om      | 50000.00 |     20 | 2020-10-11  |       101 |
|   106 | Malhar  | 65000.00 |     30 | 2022-09-09  |      NULL |
|   112 | Arjun   | 63800.00 |     10 | 2025-06-13  |      NULL |
+-------+---------+----------+--------+-------------+-----------+
7 rows in set (0.00 sec)

mysql> select E.EmpName as Employee, M.EmpName as Manager from Emp E left join Emp M on E.ManagerID = M.EmpID;
+----------+---------+
| Employee | Manager |
+----------+---------+
| Sakshi   | NULL    |
| Shruti   | Sakshi  |
| Sumit    | Sakshi  |
| Pranav   | Sakshi  |
| Om       | Sakshi  |
| Malhar   | NULL    |
| Arjun    | NULL    |
+----------+---------+
7 rows in set (0.00 sec)

mysql> select D1.DeptName as Department1,
    -> D1.DeptName as Department2,
    -> D1.DeptName as Department2,;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''
 at line 3
mysql> select D1.DeptName as Department1,
    -> D2.DeptName as Department2;
ERROR 1109 (42S02): Unknown table 'D1' in field list
mysql> SELECT
    ->     D1.DeptName AS Department1,
    ->     D2.DeptName AS Department2,
    ->     D1.Location
    -> FROM Dept D1
    -> JOIN Dept D2
    -> ON D1.Location = D2.Location
    -> AND D1.DeptID < D2.DeptID;
+-------------+-------------+----------+
| Department1 | Department2 | Location |
+-------------+-------------+----------+
| IT          | Sales       | Mumbai   |
```

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```
mysql> select D1.DeptName as Department1,
    -> D1.DeptName as Department2,
    -> D1.DeptName as Department2,;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for
 at line 3
mysql> select D1.DeptName as Department1,
    -> D2.DeptName as Department2;
ERROR 1109 (42S02): Unknown table 'D1' in field list
mysql> SELECT
    ->     D1.DeptName AS Department1,
    ->     D2.DeptName AS Department2,
    ->     D1.Location
    -> FROM Dept D1
    -> JOIN Dept D2
    -> ON D1.Location = D2.Location
    -> AND D1.DeptID < D2.DeptID;
+-------------+-------------+----------+
| Department1 | Department2 | Location |
+-------------+-------------+----------+
| IT          | Sales       | Mumbai   |
+-------------+-------------+----------+
1 row in set (0.00 sec)

mysql> select P1.ProjectName as Project1,
    -> P2.ProjectName as Project2,
    -> P1.DeptID from Project P1 join Project P2 on P1.DeptID = P2.DeptID and P1.ProjectID < P2.ProjectID;
Empty set (0.00 sec)

mysql>
```